

SMART INDIA HACKATHON 2026

- Problem Statement ID – **SIH26100**
- Problem Statement Title- **AI-Powered Integrated Bid Compliance Verification Platform For GeM Procurement**
- Theme- **Smart Automation**
- PS Category- **Software**
- Team ID-
- Team Name- **Debug Dynasty**



PROBLEM EXISTING

-  **Requirement Fragmentation** : No two GeM tenders share a checklist
-  **Evidence Fragmentation** : One clause, many documents — and reverse
-  **Self-Contradicting Bids** : PAN, GST and OEM names that disagree
-  **Unreplayable Verdicts** : The decision cannot be rebuilt later

PROPOSED SOLUTION

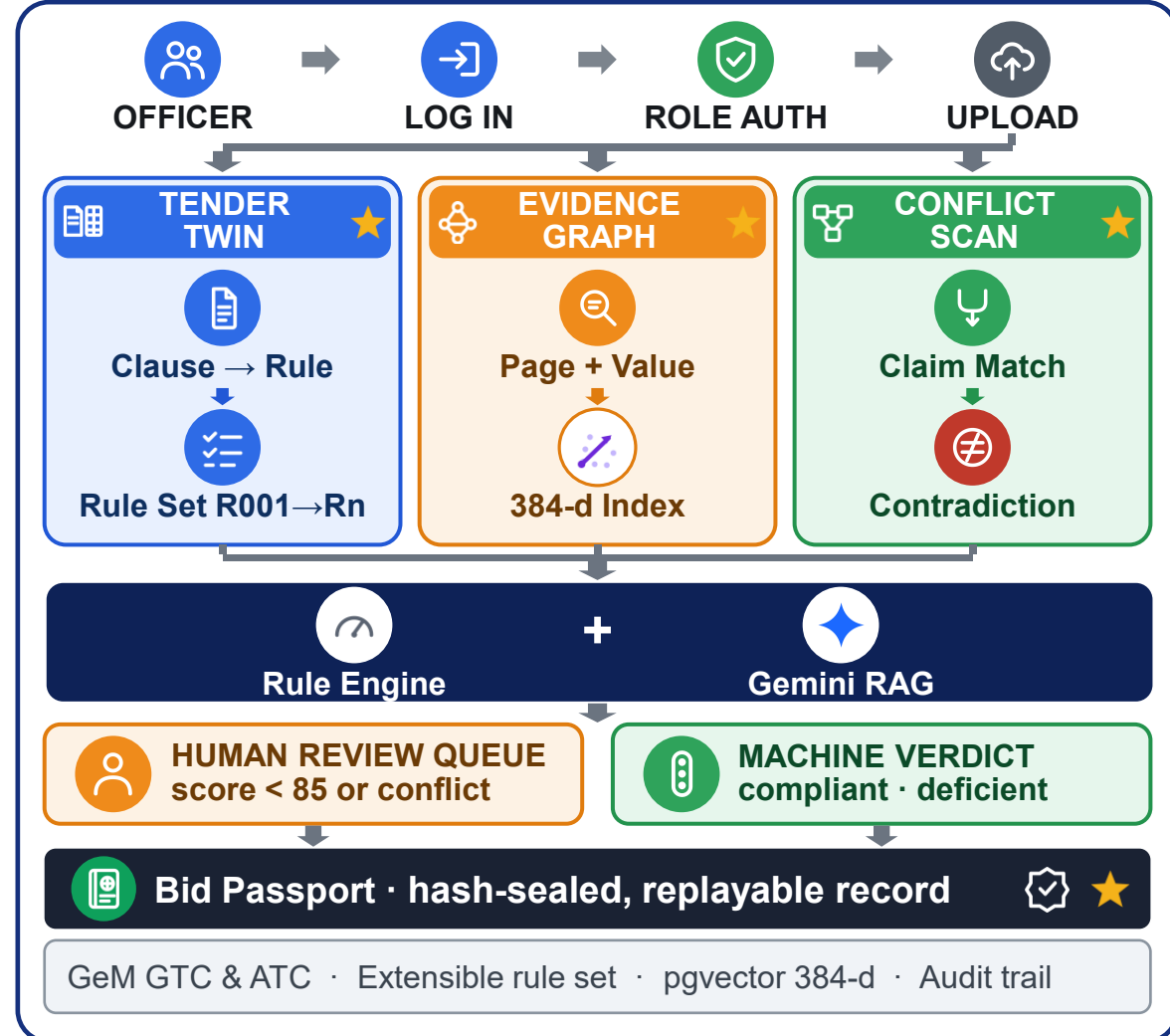
-  **Tender Digital Twin** : Clause & ATC → checkable requirement ★
-  **Evidence Graph** : Requirement → document → page → value ★
-  **Conflict Engine** : Clusters claims, flags what contradicts ★
-  **Confidence Gate** : Weak cases escalate, never auto-reject ★
-  **Bid Passport** : Hash-sealed record of the whole verdict ★

UVP (UNIQUE VALUE PROPOSITION)

“We don’t just verify the bid — we prove the decision.”

- ★ **No Evidence → No Decision** : An uncited machine claim is refused
- ★ **Tender-Aware, Not Checklist-Aware** : ATCs actually bind
- ★ **Contradiction-First** : AI hunt claims that fight each other
- ★ **Replayable Decisions** : Same evidence, same verdict, months on

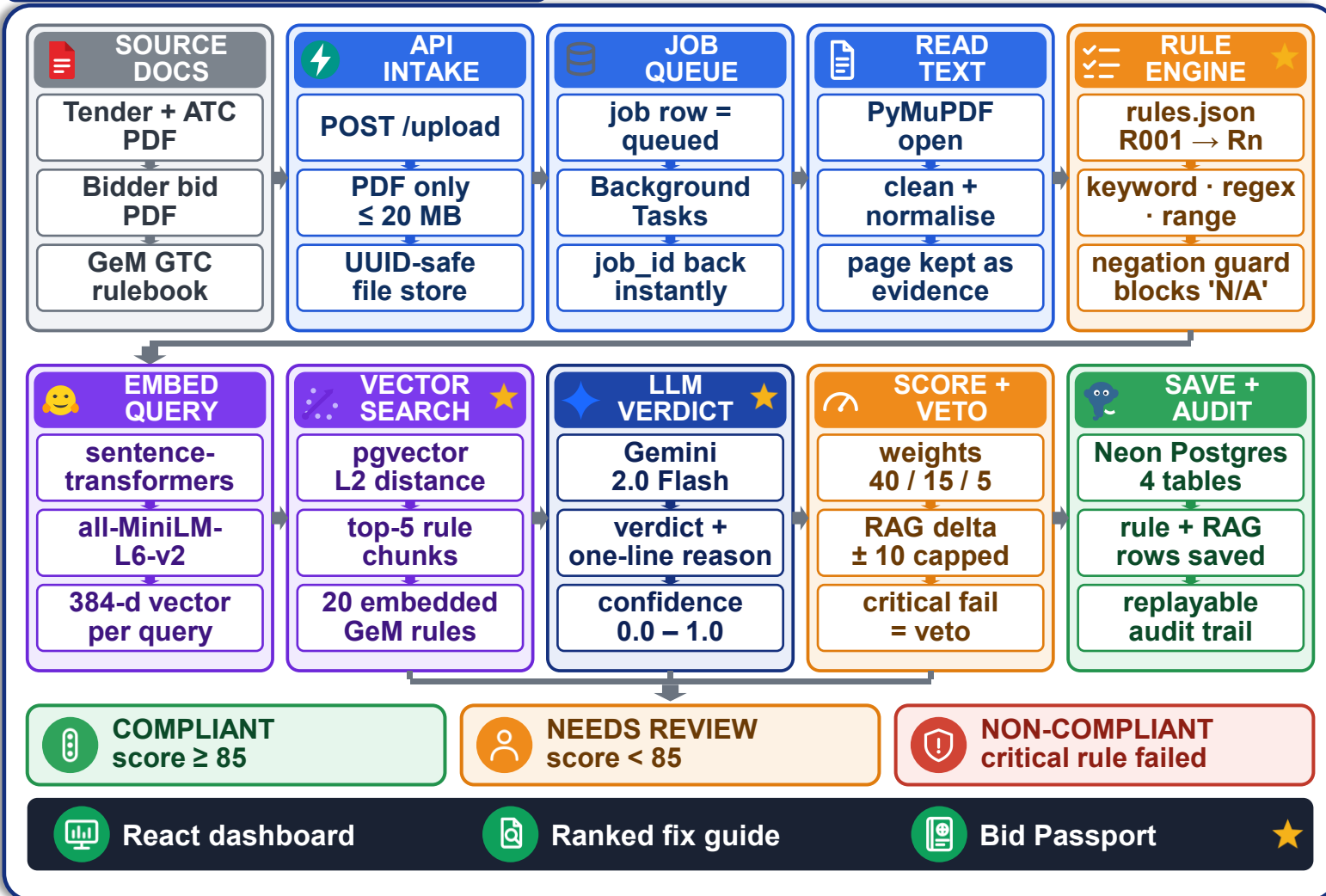
ARCHITECTURE



RECOMMENDED TECH STACK

LAYER	TECHNOLOGY	MAIN PURPOSE
Frontend	 React 19 + Vite	Upload, status, results view
API Layer	 FastAPI + Uvicorn	4 REST routes, CORS, auto docs
Async Jobs	 Background Tasks	Non-blocking run, job_id at once
PDF Parsing	 PyMuPDF	Per-page text + page number
Rule Engine	 Python re + rules.json	Rule checks with severity tags
Validators	 GSTIN · PAN · EMD regex	Format-true, not keyword-true
Embeddings	 all-MiniLM-L6-v2	384-d sentence vectors, local
Vector Search	 pgvector Vector(384)	L2 top-5 rulebook retrieval
LLM Layer	 Gemini 2.0 Flash	Grounded verdict + confidence
Database	 Neon Postgres	4 tables, full audit trail
ORM	 SQLAlchemy 2.0	Declarative models + sessions
Secrets	 python-dotenv	Keys stay out of the codebase

ARCHITECTURE OF PROTOTYPE



★ - Innovation

FEASIBILITY

Technical: Dual-layer verification — instant Rule Engine checks backed by Gemini-powered RAG for complex cases.

Innovation: Self-aware fallback — the system knows when it's unsure and auto-escalates to AI reasoning.

Operational: One-click simplicity — upload a bid, get it verified, zero manual routing.

Economic: AI fires only when truly needed — maximum accuracy at minimum compute cost.

VIABILITY

Market: Digital procurement is booming — fast, reliable bulk verification remains unsolved.

Policy & Value: Turns compliance checking into a transparent, auditable process.

Investment & ROI: Lower manual effort, faster turnaround, compounding productivity gains.

Scalability & Adoption: New rules added in minutes; AI absorbs the rest — grows without a rebuild.

TECHNICAL CHALLENGES

Risk: Rigid rules can misfire on complex wording; AI can waver on ambiguous clauses.

Mitigation: Hybrid fallback — Rule Engine first, RAG + Gemini AI for deeper contextual checks.

BUSINESS / USER CHALLENGES

Risk: Users may hesitate to trust an automated system for compliance decisions.

Mitigation: Every verdict comes with clear reasoning, a confidence score, and supporting evidence.

● **Faster & Consistent Bid Evaluation**

Automated requirement-vs-document checking replaces line-by-line manual review, and one shared rule set makes different officers reach the same verdict on the same clause.

● **Transparent & Defensible Procurement**

Every flagged requirement links to its source document, extracted value and page — a stated reason, not an unexplained AI score, so decisions survive audit and appeal.

● **Reduced Compliance & Fraud Risk**

Cross-document validation catches missing evidence, conflicting bidder details and expired certificates early; low-confidence cases go to a reviewer, not an auto-rejection.

Feature	Existing Solutions	GeMShield
Tender-specific rule generation	✗	✓
Cross-document verification	✗	✓
Clause → evidence → verdict trail	✗	✓
Hybrid rules + RAG AI	⚠	✓
GeM procurement logic built in	✗	✓
Negation-aware field validation	✗	✓
Confidence-gated human review	⚠	✓
On-premise / data sovereignty	⚠	✓
Reusable, auditable decision record	✗	✓

HOW GEMSHIELD WORKS

MODE 1 : RULE ENGINE — clear clause

For : Exactly checkable requirements

Process : Extract → GeM rules → verdict

Time : Under 2 seconds

Confidence : Deterministic · repeatable

Best For : GSTIN, PAN, EMD, dates, values

MODE 2 : RAG + AI — unclear clause

For : Wording no rule reads literally

Process : Embed → pgvector → Gemini

Time : 3 – 8 seconds

Confidence : Rulebook-grounded · scored

Best For : Paraphrased or conditional ATCs

MODE 3 : HUMAN REVIEW — low score

For : Conflicting or low-confidence

Process : Flag + evidence → officer decides

Time : Officer-led, fully logged

Confidence : Officer has final authority

Best For : Contradictions, poor scans

RESEARCH FOUNDATION

- **General Financial Rules (GFR) 2017** define India's statutory eligibility & EMD thresholds. We also studied the **CVC Integrity Guidelines** for conflict-of-interest checks. *Full details:* [GFR 2017 \(Dept. of Expenditure\)](#) / [CVC Guidelines](#)
- **Industry studies on RegTech** show OCR + NLP document analysis can flag missing certificates before manual review. *Links:* [Minaions](#) / [Bidz365](#)
- **GeM market data** shows a scale of ₹20 lakh crore GMV and 3.78 crore orders (Aug 2026) that manual review can't keep up with. *Links:* [PIB GeM Update](#) / [GeM Portal](#)

SOLUTIONS ALREADY EXIST

- Most existing tools check seller eligibility pre-bid — none verify the buyer's compliance checklist.
- **Existing tools:** [Minaions](#) / [Bidz365](#) / [Tenderkart](#) / [RITES](#)

WHY GeMSHIELD STANDS OUT

- ✓ **Hybrid AI:** Rules stay auditable, explainable
- ✓ **Document-level:** Verifies GST, PAN, MSME, OEM & EMD
- ✓ **Buyer-side:** Built for the evaluator, not the seller
- ✓ **Explainable:** Flags cite the exact clause & mismatch